

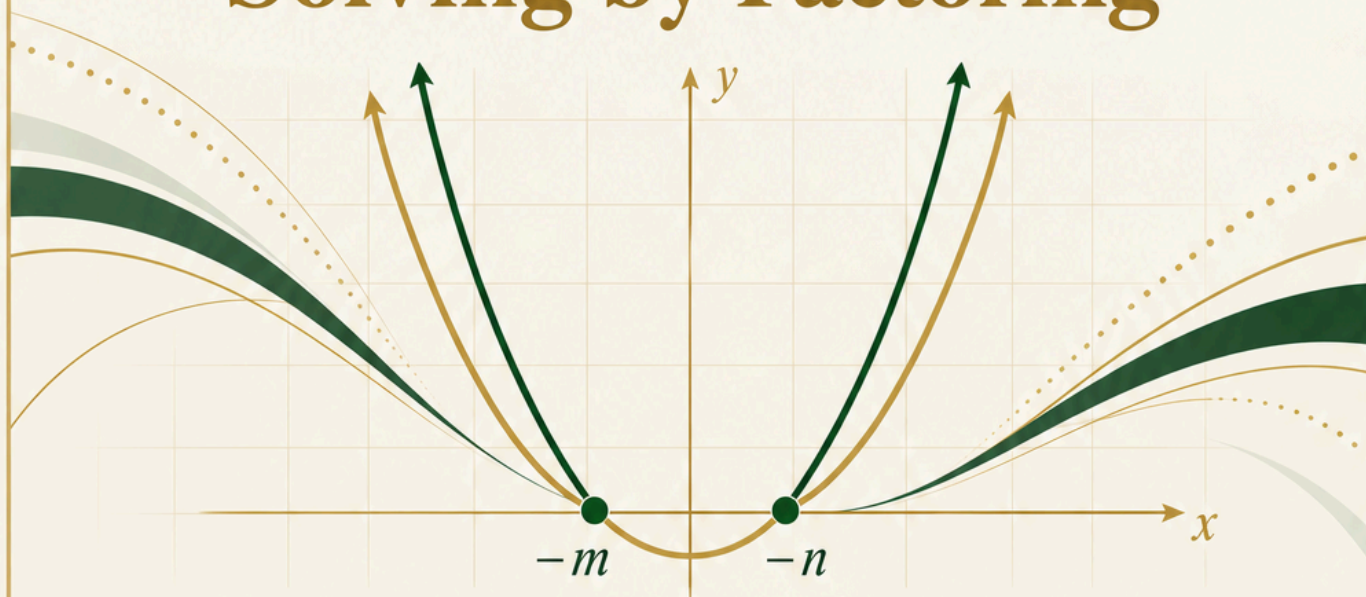


ALGEBRA 1

CLASS 14

Introduction to Quadratic Equations

Solving by Factoring



$$x^2 + bx + c = (x + m)(x + n)$$

$$m + n = b \quad \bullet \quad mn = c$$

Recognize patterns. Factor confidently. Find solutions.

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1. Recognizing Quadratic Equations

A quadratic equation has a highest variable power of 2.

$$ax^2 + bx + c = 0$$
$$a \neq 0$$

a: coefficient of x^2 | ***b***: coefficient of x | ***c***: constant

EXAMPLE

$$x^2 + 5x + 6 = 0$$
$$a = 1 \quad b = 5 \quad c = 6$$

QUADRATIC

$$x^2 - 4x = 0$$

Highest power: **2**

LINEAR

$$3x + 6 = 0$$

Highest power: **1**

NOTE: | Simplify first, then identify the highest power.

Image 1 practice

Recognizing quadratic equations

Q1. Is $x^2 + 8x + 12 = 0$ quadratic? Identify a, b, and c.

Q2. Classify $5x - 18 = 0$ as linear or quadratic. Explain.

Q3. Is $4x^2 - 36 = 0$ quadratic even though there is no x term? Give a, b, and c.

Q4. Simplify $3x^2 + 2x = x^2 + 10$ into standard form, then identify a, b, and c.

Q5. Is $x(x + 6) = x^2 + 18$ quadratic after simplification? Solve it.

2. Writing an Equation in Standard Form

Move all terms to one side and combine like terms.

$$ax^2 + bx + c = 0$$

$$a \neq 0$$

STEP 1

Start with the equation.

$$x^2 + 5x = 6$$

STEP 2

Subtract 6 from both sides.

$$x^2 + 5x - 6 = 6 - 6$$

STEP 3

Simplify.

$$x^2 + 5x - 6 = 0$$

Now the equation is in standard form.

COMBINE LIKE TERMS

$$x^2 + 2x + 3 = 7x + 9$$

$$\downarrow$$

$$x^2 + 2x + 3 - 7x - 9 = 0$$

$$\downarrow$$

$$x^2 - 5x - 6 = 0$$



Keep the equation balanced: perform the same operation on both sides.

Image 2 practice

Writing equations in standard form

Q6. Write $x^2 + 7x = 18$ in standard form. Do not solve.

Q7. Write $x^2 = 4x + 21$ in standard form. Do not solve.

Q8. Write $2x^2 + 5x + 3 = x + 15$ in standard form. Do not solve.

Q9. Expand and write $x(x + 5) = 24$ in standard form. Do not solve.

Q10. Expand both sides of $(x + 2)(x + 4) = 3x + 26$ and write standard form.

3. The Zero-Product Property

If a product equals zero, at least one factor must equal zero.

If $A \times B = 0$, then $A = 0$ or $B = 0$.

$$0 \times 7 = 0$$

$$5 \times 0 = 0$$

SET EACH FACTOR EQUAL TO ZERO

$$(x - 2)(x + 3) = 0$$

$$x - 2 = 0$$

$$x + 3 = 0$$

$$x = 2$$

$$x = -3$$

Solutions: $x = 2$ or $x = -3$.

CHECK

For $x = 2$: $(2 - 2)(2 + 3) = 0 \times 5 = 0$

For $x = -3$: $(-3 - 2)(-3 + 3) = (-5) \times 0 = 0$

Key Reminder: Apply this property only when the product equals zero.

Image 3 practice

Using the zero-product property

Q11. Use the zero-product property to solve $(x - 7)(x + 4) = 0$.

Q12. Solve $x(x + 9) = 0$. Do not divide by x .

Q13. Solve $(2x - 5)(x + 1) = 0$. Keep fractional answers exact.

Q14. Solve $(3x + 2)(2x - 7) = 0$.

Q15. A student sets each factor to zero in $(x - 4)(x + 2) = 16$. Explain why that is invalid, then solve correctly.



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4. Factoring Out the GCF

GCF = Greatest Common Factor

Factor out the GCF, then set each factor equal to zero.

STEP 1 • Identify the common factor

$$x^2 - 4x = 0$$

Both terms contain x .
The GCF is x .

STEP 2 • Factor

$$x(x - 4) = 0$$

STEP 3 • Use the zero-product property

$$x = 0 \quad \text{OR} \quad x - 4 = 0$$
$$x = 4$$

Solutions: $x = 0$ or $x = 4$

WHEN THE GCF INCLUDES A NUMBER

$$2x^2 + 6x = 0$$

$$2x(x + 3) = 0$$

$$x = 0 \text{ or } x = -3$$

The constant factor 2
cannot equal zero.

IMPORTANT TIP

Do not divide by x : you could lose
the solution $x = 0$.

Image 4 practice

Factoring out the greatest common factor

Q16. Solve $x^2 - 7x = 0$ by factoring. Check every solution.

Q17. Solve $x^2 + 11x = 0$ by factoring. Check every solution.

Q18. Solve $3x^2 - 18x = 0$ by factoring. Check every solution.

Q19. Solve $4x^2 + 10x = 0$ by factoring. Check every solution.

Q20. Solve $6x^2 = 15x$ by factoring. Check every solution.

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5. Solving by Factoring Trinomials

For $x^2 + bx + c = 0$, find two numbers whose product is c and whose sum is b .

1 STEP 1 • Find the two numbers

$$x^2 + 5x + 6 = 0$$

Product:
 $2 \times 3 = 6$

Sum:
 $2 + 3 = 5$

2 STEP 2 • Factor the trinomial

$$(x + 2)(x + 3) = 0$$

3 STEP 3 • Set each factor equal to zero

$$\begin{aligned} x + 2 &= 0 \\ x &= -2 \end{aligned}$$

$$\begin{aligned} x + 3 &= 0 \\ x &= -3 \end{aligned}$$

Solutions: $x = -2$ or $x = -3$

TRY DIFFERENT SIGNS

$$x^2 - x - 6 = 0$$

Product: $(-3)(2) = -6$ | Sum: $-3 + 2 = -1$

$$(x - 3)(x + 2) = 0$$

Solutions: $x = 3$ or $x = -2$



Check the signs when solving each factor.

Image 5 practice

Solving trinomials by factoring

Q21. Solve $x^2 + 9x + 20 = 0$ by factoring. Check every solution.

Q22. Solve $x^2 - 11x + 28 = 0$ by factoring. Check every solution.

Q23. Solve $x^2 + x - 30 = 0$ by factoring. Check every solution.

Q24. Solve $x^2 - 2x = 35$ by factoring. Check every solution.

Q25. Solve $2x^2 + 7x + 3 = 0$ by factoring. Check every solution.

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6. Solving Special Cases

Recognize patterns to factor efficiently.

1 DIFFERENCE OF SQUARES

Pattern:

$$A^2 - B^2 = (A - B)(A + B)$$

Example: Solve $x^2 - 9 = 0$.

$$\begin{aligned} x^2 - 3^2 &= 0 \\ (x - 3)(x + 3) &= 0 \end{aligned}$$

$\swarrow$
 $x - 3 = 0$
 $\downarrow$
 $x = 3$

$\searrow$
 $x + 3 = 0$
 $\downarrow$
 $x = -3$

Two distinct solutions: $x = 3$ or $x = -3$.

2 PERFECT-SQUARE TRINOMIAL

Pattern:

$$A^2 - 2AB + B^2 = (A - B)^2$$

Example: Solve $x^2 - 6x + 9 = 0$.

$$\begin{aligned} (x - 3)^2 &= 0 \\ (x - 3)(x - 3) &= 0 \\ x - 3 &= 0 \\ \downarrow \\ x &= 3 \end{aligned}$$

One distinct solution: $x = 3$.

A repeated root (multiplicity 2).



Factor first. Then apply the zero-product property.

Image 6 practice

Solving special cases

Q26. Solve $x^2 - 49 = 0$ by factoring. Check every solution.

Q27. Solve $x^2 + 10x + 25 = 0$ by factoring. Check every solution.

Q28. Solve $4x^2 - 81 = 0$ by factoring. Check every solution.

Q29. Solve $3x^2 - 48 = 0$ by factoring. Check every solution.

Q30. Solve $4x^2 - 20x + 25 = 0$ by factoring. Check every solution.

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7. Checking Solutions & Avoiding Mistakes

CHECK IN THE ORIGINAL EQUATION

$$x^2 + 5x + 6 = 0$$

Solutions: $x = -2$ or $x = -3$

$$x = -2: \quad (-2)^2 + 5(-2) + 6 = 4 - 10 + 6 = 0$$

$$x = -3: \quad (-3)^2 + 5(-3) + 6 = 9 - 15 + 6 = 0$$

Both solutions satisfy the original equation.

1

Set the equation equal to zero

The zero-product property requires a product equal to zero.

$$x^2 + 5x = 6 \rightarrow x^2 + 5x - 6 = 0$$

2

Keep the signs correct

$$x + 3 = 0 \rightarrow x = -3$$

$$x - 3 = 0 \rightarrow x = 3$$

3

Do not divide by x

$$x^2 - 4x = 0 \rightarrow x(x - 4) = 0$$

Solutions: $x = 0$ or $x = 4$

Dividing by x would lose the solution $x = 0$.

Standard form $\rightarrow$ Factor $\rightarrow$ Solve each factor $\rightarrow$ Check

Image 7 practice

Checking solutions and avoiding mistakes

Q31. For $x^2 - 3x - 28 = 0$, test $x = 7$ and $x = -4$. Do both work?

Q32. A student factors $x^2 + 2x - 15 = 0$ as $(x + 5)(x - 3) = 0$, then reports 5 and -3. Correct the solutions and check.

Q33. A student divides $5x^2 - 30x = 0$ by x and finds only $x = 6$. Find the missing solution and explain.

Q34. Mia changes $x^2 - 8x = 9$ into $x^2 - 8x + 9 = 0$. Correct the sign, factor, and check.

Q35. Noah claims $x = 2$ or $x = -5$ solves $(x - 2)(x + 5) = 18$. Explain the mistake and solve the original equation.

8. Solving Quadratic Word Problems

WORD PROBLEM

A rectangular garden has an area of 40 m^2 . Its length is 3 m greater than its width. Find its dimensions.



1 STEP 1 • Define and model

Let x be the width in meters.

$$\text{Length} = x + 3$$

$$x(x + 3) = 40$$

2 STEP 2 • Write standard form and factor

$$x^2 + 3x - 40 = 0$$

$$(x + 8)(x - 5) = 0$$

3 STEP 3 • Solve and interpret

$$x = -8 \text{ or } x = 5$$

Reject -8 : a garden cannot have a negative width.

$$\begin{aligned} \text{Width} &= 5 \text{ m} \\ \text{Length} &= 8 \text{ m} \end{aligned}$$

CHECK

$$5 \times 8 = 40 \text{ m}^2$$

$$8 - 5 = 3 \text{ m}$$

A mathematical solution must also make sense in the real situation.

Image 8 practice

Solving quadratic word problems

Q36. A square mosaic has area 64 square inches. Let x be its side length. Write and solve an equation to find x .

Q37. A rectangular art board has length 2 cm more than its width and area 48 cm^2 . Find its dimensions.

Q38. Two consecutive positive integers have product 90. Find both integers.

Q39. A display is 5 feet longer than it is wide. Its area is 84 square feet. Find the width and length.

Q40. A photo is twice as long as it is wide. Its area is 72 cm^2 . Find both dimensions.

Homework review

R1–R5

R1. Image 1: Simplify $5x^2 - 2x = 2x^2 + 12$ into standard form. Is it quadratic? Identify a, b, and c.

R2. Image 2: Write $(x + 1)(x + 5) = 2x + 17$ in standard form without solving.

R3. Image 3: Solve $(2x + 3)(x - 8) = 0$ using the zero-product property.

R4. Image 4: Solve $6x^2 + 21x = 0$ by factoring. Check every solution.

R5. Image 5: Solve $x^2 - 3x - 40 = 0$ by factoring. Check every solution.

Homework review

R6–R8

R6. Image 6: Solve $9x^2 - 24x + 16 = 0$ by factoring. Check every solution.

R7. Image 7: A student claims $x = 7$ and $x = -3$ solve $x^2 + 4x = 21$. Correct both signs and check.

R8. Image 8: A rectangular garden is 6 m longer than it is wide and has area 72 m^2 . Find both dimensions.

Homework word problems

W1–W5

W1. A square tile has area 121 cm^2 . Find its side length.

W2. A rectangular rug is 1 m longer than it is wide. Its area is 30 m^2 . Find its dimensions.

W3. A rectangular tabletop has length 3 feet more than its width and area 54 square feet. Find its dimensions.

W4. Two consecutive positive integers have product 132. What are they?

W5. Two positive even integers differ by 2. Their product is 168. Find them.

Homework word problems

W6–W10

W6. Chairs form a rectangular array. There are 4 more chairs in each row than there are rows, and 96 chairs total. Find the number of rows and chairs per row.

W7. A rectangle is twice as long as it is wide. Its area is 98 square meters. Find its dimensions.

W8. A number squared is 6 more than 5 times the number. Find every number that fits the description.

W9. A poster is 3 cm longer than twice its width. Its area is 65 cm^2 . Find its width and length.

W10. A square patio is enlarged by adding 2 m to each side length. The enlarged patio has area 81 m^2 . Find the original side length and original area.

Homework word problems

W11–W12

W11. A 6 m by 10 m garden has a path of uniform width x around its outside. The combined garden-and-path area is 96 m^2 . Find the path width.

W12. A club charges \$20 per ticket and sells 30 tickets. For each \$1 reduction, it sells 2 additional tickets. Let x be the number of \$1 reductions. What ticket prices would produce \$608 in revenue? Find all valid prices.